Assignment Brief

COM6012 Scalable Machine Learning 2026 Assignment

Deadline: 13:00 Wednesday 6th May 2026

Please carefully read the assignment brief before starting to complete the assignment.
Release Status: complete

[Q1 - 10 marks](#)
[Q2 - 10 marks](#)
[Q3 - 10 marks](#)
[Q4 - 10 marks](#)

An [FAQ](#) will be updated when questions arise about important clarifications or tips.

Assignment Brief

How and what to submit

A. Create a folder **YOUR_USERNAME-COM6012-2026** containing the following:

1) **AS_report.pdf**: A report in **PDF format containing answers (including all figures and tables) to ALL questions** at the root of the zipped folder (like readme.txt in the lab solutions). If an answer to a question is not found in this PDF file, you will lose the respective mark. The report should be concise. You may include appendices/references for additional information, but marking will focus on the main body of the report.
2) **Code, batch scripts, and output files (see the sample solutions to the lab exercises as examples): All files (e.g., .py, .sh, .txt, .png, .csv) used to generate the answers to the** individual questions in the report above, **except the data**, should be included. These files should be named properly, starting with the question number (separate files for the two questions): **for example**, your Python code file should be named as **Q1_code.py and Q2_code.py, your HPC scripts as Q1_script.sh and Q2_script.sh**, and your output files from the HPC cluster as **Q1_output.txt and Q2_output.txt** (and Q1_C1_fig.jpg, etc.). The **results must be generated on the HPC cluster, not your local machine**. **Figures must be created by Python code**. We will apply a **25% penalty against the mark for a question for each missing file (code, batch script, output file, report)**.

Example folder structure:

```Shell
YOURUSERNAME-COM6012-2026/
|
├── AS_report.pdf
|
├── Q1/
|    ├── Q1_code.py
|    ├── Q1_script.sh
|    ├── Q1_output.txt
|    ├── Q1_fig1.png
|    ├── Q1_***.csv (generated by your code, not the input data)
|    └── ...
|
├── Q2/
|    ├── Q2_code.py
|    ├── Q2_script.sh
|    ├── Q2_output.txt
|    └── ...
|
├── Q3/
|    ├── Q3_code.py
|    └── ...
|
└── Q4/
|    ├── Q4_code.py
|    └── ...
└── ...
```

B. When you have finished ALL of the questions, zip your folder **YOUR_USERNAME-COM6012-2026** to include all of the files described above (one single report plus code, script, and output files for all questions, properly named) and upload this **YOUR_USERNAME-COM6012-2026.zip** file to Blackboard before the deadline. Double-check that these files are included by downloading the zipped folder from Blackboard to another machine and opening it.

C. **NO DATA UPLOAD**: Please do not upload the data files used. Instead, use the relative file path in your code, assuming data files are downloaded (and unzipped if needed) **under the folder 'Data'**, as in the lab.

D. **Code and output**: 1) Use **PySpark 4.1.0 and Python 3.13** as covered in the lecture and lab sessions to complete the tasks; 2) **Submit your PySpark job to HPC** with **sbatch** to obtain the output. Output files must be produced by the HPC batch jobs submitted via sbatch.

Assessment Criteria (Scope: Sessions 1 to 8; Total: 40 marks - 10 marks/question)

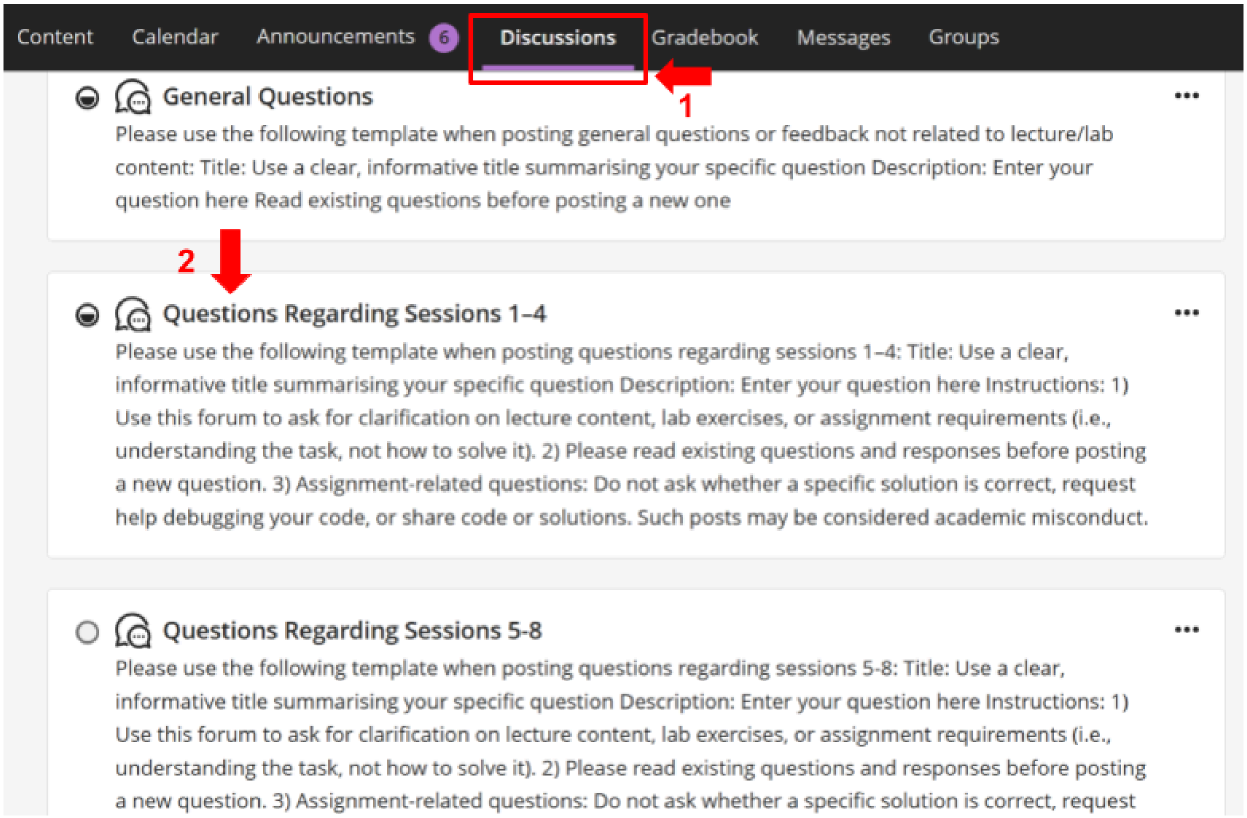
1. Being able to use PySpark to analyse big data to answer data analytics questions.
2. Being able to perform tasks covered in Sessions 1 to 8 on large-scale data.
3. Being able to make useful observations and clearly explain the obtained results.

Late submissions: We follow the university's guidelines about late submissions, i.e., “If you submit a piece of coursework late, your mark will be deducted by **5% for each working day after the deadline**.
If you submit more than five working days after the deadline, **you will receive a mark of zero**.”
Please see this link.
Extenuating circumstances: If you are experiencing difficulties that are affecting your assignment submission, you may be able to apply for extenuating circumstances so these difficulties can be taken into account. Please see this link.
Academic misconduct: "The consequences of academic misconduct can be serious. You may face action within your school or at a University discipline hearing." Please carefully read this link on different types of Academic Misconduct and how to avoid them if you are not sure.

If you have any questions, please post them on the Blackboard discussion board by following the steps below:

1. Click Discussion on the top panel
2. Select the discussion relevant to your question (Sessions 1–4 or Sessions 5–8)
3. Read the instructions carefully and post your question using the provided template

Please note: As mentioned in the first lecture, we **do not** answer questions about the assignment, lectures, labs, or the exam via email.


Content
Calendar
Announcements
6
Discussions
Gradebook
Messages
Groups
1
General Questions
Please use the following template when posting general questions or feedback not related to lecture/lab content: Title: Use a clear, informative title summarising your specific question Description: Enter your question here Read existing questions before posting a new one
2
Questions Regarding Sessions 1–4
Please use the following template when posting questions regarding sessions 1–4: Title: Use a clear, informative title summarising your specific question Description: Enter your question here Instructions: 1) Use this forum to ask for clarification on lecture content, lab exercises, or assignment requirements (i.e., understanding the task, not how to solve it). 2) Please read existing questions and responses before posting a new question. 3) Assignment-related questions: Do not ask whether a specific solution is correct, request help debugging your code, or share code or solutions. Such posts may be considered academic misconduct.
Questions Regarding Sessions 5-8
Please use the following template when posting questions regarding sessions 5-8: Title: Use a clear, informative title summarising your specific question Description: Enter your question here Instructions: 1) Use this forum to ask for clarification on lecture content, lab exercises, or assignment requirements (i.e., understanding the task, not how to solve it). 2) Please read existing questions and responses before posting a new question. 3) Assignment-related questions: Do not ask whether a specific solution is correct, request

Discussion Topic

Follow

Please use the following template when posting questions regarding sessions 1–4:

Title: Use a clear, informative title summarising your specific question

Description: Enter your question here

Instructions:

1) Use this forum to ask for clarification on lecture content, lab exercises, or assignment requirements (i.e., understanding the task, not how to solve it).

2) Please **read existing questions and responses before** posting a new question.

3) Assignment-related questions: **Do not** ask whether a specific solution is correct, request help debugging your code, or share code or solutions. Such posts may be considered **academic misconduct**.

3

Type a Post

Post

Shuo Zhou INSTRUCTOR

6 Feb 2026 11:49 (Edited by Shuo Zhou on 6 Feb 2026 11:53)

Example post

A question about lecture 1.

Question 1

Question 1. Log Mining and Analysis [10 marks, set by Shuo Zhou]

Please carefully read the assignment brief before starting to complete the assignment.

Prerequisites:
Before attempting this question, you are expected to have completed Lab 1 and Lab 2.

Data:

Download the NASA access log for July 1995 using:

```Shell
wget ftp://ita.ee.lbl.gov/traces/NASA_access_log_Jul95.gz
```

Unzip the file, for example:

```Shell
gunzip NASA_access_log_Jul95.gz
```

Then move the file to the "Data" folder, for example,
'`/users/your_username/com6012/ScalableML/Data`'.

Definitions

For this assignment:

- A request = one log entry.
- A unique host = a unique hostname appearing in the host field.
- A UK academic host = hostname ending with .ac.uk
- A UK company host = hostname ending with .co.uk
- A UK government host = hostname ending with .gov.uk
- For institution/company level: Use the last three labels of the hostname (e.g., shef.ac.uk, cam.ac.uk, bt.co.uk, nhs.gov.uk).

Tasks:

A. For each of the following categories:

- .ac.uk (UK academic institutions)
- .co.uk (UK companies)
- .gov.uk (UK government departments)

Compute:

- Total number of requests

- Number of unique hostnames

Present your results as a table in your report with columns:

```Shell
| Sector  | Total Requests | Unique Hosts |
|---------|----------------|--------------|
| .ac.uk  |                |              |
| .co.uk  |                |              |
| .gov.uk |                |              |
```

[1 mark]

B. Using UK academic hosts (*.ac.uk):

1. Compute the total number of requests from the University of Sheffield domain (ending with shef.ac.uk).
2. Find out how many other UK academic institutions (e.g., cam.ac.uk, ox.ac.uk) made more requests than Sheffield.
3. Create a DataFrame to store the above institutions with more requests than Sheffield, along with the number of requests for each. Then save the DataFrame as a CSV file.

You are expected to explain:

- How do you extract a “UK academic institution domain”
- How do you aggregate requests at the institution level

To demonstrate this, you may include your code segments and the resulting table in your report.

[2 marks]

C. Using UK company hosts (*.co.uk):

- Identify the 9 most active UK companies by total number of requests.
- Produce a visualisation showing the number of requests for each of the top 9 companies compared with all the other companies (10 items in total).

You are free to choose an appropriate visualisation type. Your figure must:

- be clearly labelled
- be interpretable with minimum explanation (caption)
- be created by your Python program and saved as a .png/.jpg file

Briefly justify your design choice and include the figure in your report.

[2 marks]

D. Create heatmaps showing access patterns for:

1. The most active UK academic institution domain (*.ac.uk)
2. The most active UK company domain (*.co.uk)
3. All UK government domains combined (gov.uk)

(“Most active” = highest total number of requests.)

For each heatmap:

- x-axis: Day of month (1-28)
- y-axis: Hour of day (0–23)
- Colour intensity: Number of requests

Produce three separate heatmaps using your Python program and save them as .png/.jpg files. These figures should be included in your report.

[3 marks]

E. Select two observations from Tasks A–D that you find interesting.
For each observation, write at least three sentences addressing the following questions:

1. What is the observation?
2. What are the possible causes?
3. How could this insight be useful to NASA?

Each observation must refer to specific results or figures from your analysis.

[2 marks]

Question 2

Question 2. Scalable Logistic Regression and Generalized Linear Models [10 marks, set by Shuo Zhou]

Please carefully read the assignment brief before starting to complete the assignment.

Prerequisites:
Before attempting this question, you are expected to have completed Lab 3 and Lab 4.

Data:

You are given a UK traffic vehicle-count dataset containing over 5 million observations collected from 43,105 counting points from 2000 to 2024. Each record corresponds to the number of vehicles recorded at a traffic counting point during a specific hour.

This dataset has been uploaded to Stanage and is available at:

```shell
/mnt/parscratch/users/com6012_2026/data/dft_traffic_counts_raw_counts.csv
```

You have read access to this directory, but you do not have write permission.

For more information about the dataset, see the government website:
https://roadtraffic.dft.gov.uk/downloads#road-traffic-data-guidance.

Task:

Use 10 cores and 10 GB of memory per core for the following tasks. The entire program should take approximately 1 hour to complete, depending on your implementation. Cache intermediate DataFrames when appropriate to avoid recomputing expensive transformations.

A. Load and process the dataset

Load the dataset from the CSV file into a Spark DataFrame. The dataset contains the following columns:

```python
['count_point_id', 'direction_of_travel', 'year', 'count_date', 'hour',
```

```
'region_id', 'region_name', 'region_ons_code', 'local_authority_id',
'local_authority_name', 'local_authority_code', 'road_name',
'road_category', 'road_type', 'start_junction_road_name',
'end_junction_road_name', 'easting', 'northing', 'latitude',
'longitude', 'link_length_km', 'link_length_miles', 'pedal_cycles',
'two_wheeled_motor_vehicles', 'cars_and_taxis', 'buses_and_coaches',
'LGVs', 'HGVs_2_rigid_axle', 'HGVs_3_rigid_axle',
'HGVs_4_or_more_rigid_axle', 'HGVs_3_or_4_articulated_axle',
'HGVs_5_articulated_axle', 'HGVs_6_articulated_axle', 'all_HGVs',
'all_motor_vehicles']
```

Perform the following preprocessing steps:

1. Create two new columns, 'month' and 'weekday', from the 'count_date' column. You may find the following functions useful:

- weekday: https://spark.apache.org/docs/4.1.0/api/python/reference/pyspark.sql/api/pyspark.sql.functions.weekday.html
- month: https://spark.apache.org/docs/4.1.0/api/python/reference/pyspark.sql/api/pyspark.sql.functions.month.html

2. Convert all values in the column 'direction_of_travel' to uppercase

3. Drop the rows where 'all_motor_vehicles' is NULL. You may find this dropna function useful: https://spark.apache.org/docs/latest/api/python/reference/pyspark.sql/api/pyspark.sql.DataFrame.dropna.html.

4. Use the one-hot encoder (https://spark.apache.org/docs/4.1.0/api/python/reference/api/pyspark.ml.feature.OneHotEncoder.html) to
encode the categorical features:

```python
['direction_of_travel', 'hour', 'region_ons_code', 'local_authority_code',
'month', 'weekday']
```

5. Ensure the following numerical features are of type double, and **standardize** them using StandardScaler:

```python
['latitude', 'longitude']
```

6. Split the data by year as follows, and report the sample size for each set:
    - Training: 2000–2021
    - Validation: 2022–2023
    - Testing: 2024

[3 marks]

Note: ***Do not*** use any vehicle subtype counts (i.e., 'pedal_cycles', 'two_wheeled_motor_vehicles', 'cars_and_taxis', 'buses_and_coaches', 'LGVs', 'HGVs_2_rigid_axle', 'HGVs_3_rigid_axle', 'HGVs_4_or_more_rigid_axle', 'HGVs_3_or_4_articulated_axle', 'HGVs_5_articulated_axle', 'HGVs_6_articulated_axle', 'all_HGVs') as features, as these directly contribute to the target variable.

B. Poisson regression

Use the processed features:

```python
['direction_of_travel', 'hour', 'region_ons_code', 'local_authority_code',
'month', 'weekday','latitude', 'longitude']
```

to predict: 'all_motor_vehicles'.

Tune the hyperparameter regParam over [0.001, 0.01, 0.1, 1, 10, 100, 1000]. For each candidate value of regParam, create 5 random subsets by sampling 80% of the training set. Use the **last five digits** of your student registration number as the starting seed to generate five seeds. For example, if your student registration number is 111111111, use 11111, 11112, 11113, 11114, and 11115 as the random seeds. Please include your student registration number in both your code comments and your report.

For each sampled subset, train the model on that subset and evaluate it on the validation set. Then, for each value of regParam, compute the mean and standard deviation of the 5 validation MSE values. Plot the mean validation MSE against regParam, with error bars showing ± the standard deviation. Include the validation curve in your report and discuss the optimal value of regParam.

Retrain the Poisson regression model on the combined training and validation sets using the optimal regParam obtained in the previous step. Evaluate the final model on the test set and report the test MSE, along with the learned model coefficients.

Hint: Saving results to a separate file (e.g., .csv or .txt) instead of printing them, or setting the Spark log level to ERROR using setLogLevel, can make it easier to locate the results to report and reduce the size of the generated log file. The same applies to Task C.

[3 marks]

C. Logistic regression
Using the training set only, compute the median value of 'all_motor_vehicles'.

Create a new binary target column 'traffic' for all instances as follows:

- traffic = 0 (low traffic) if all_motor_vehicles <= median
- traffic = 1 (high traffic) if all_motor_vehicles > median

Use the same processed features as in Task B to train a logistic regression model to predict 'traffic'.

Tune the hyperparameters regParam and elasticNetParam over [0.001, 0.01, 0.1, 1, 10, 100, 1000] and [0.0, 0.2, 0.5, 0.8, 1.0], respectively. Following the same setting as in task B, create 5 random training subsets, each obtained by sampling 80% of the training instances.

For each combination of regParam and elasticNetParam, train the model on the sampled training subsets and evaluate it on the validation set. Compute the mean validation accuracy and standard deviation across the 5 runs.

Plot the mean validation accuracy for all combinations of regParam and elasticNetParam (e.g., a grouped line plot), with error bars representing ± the standard deviation. Include the validation curve in the report and discuss the optimal values of regParam and elasticNetParam.

Retrain the logistic regression model on the combined training and validation sets using the optimal regParam and elasticNetParam obtained in the previous step. Evaluate the final model on the testing set and report the test accuracy along with the learned model coefficients.

[3 marks]

D. Select one observation or finding from Tasks A–C that you find interesting (e.g., model performance or feature importance/coefficients) and briefly discuss it.

[1 mark]

Question 3

Question 3. Scalable Decision Trees, Random Forests, and Gradient Boosting [10 marks, set by Robert Loftin]

Prerequisites: Before attempting this question, you are expected to have completed Lab 5.

Data:

In this question, you will use supervised classification algorithms to identify Higgs bosons from particle collisions, like those produced by the Large Hadron Collider. In particular, you will use the HIGGS dataset.

Use **wget** to download the data using the direct link:
http://archive.ics.uci.edu/ml/machine-learning-databases/00280/HIGGS.csv.gz

You will then need to unzip the dataset first. For this purpose, you can use a tool like **gzip** or **unzip**.

Task:

You must use a single Slurm batch script to run your code for Q2 on Stanage for this question. Your batch script should request 10 CPU cores in total, and 8 GB of memory per core.

In part A of this question, you will apply Random Forests and Gradient Boosted Trees to a subset of the dataset, while in part B you will apply the same methods to the full dataset. To evaluate the performance of each model, you will use the binary classification accuracy.

A.

Use Spark pipelines, parameter grids, and cross-validation to find the best configuration of parameters for both the Random Forest and Gradient-Boosted Tree classification models on the HIGGS dataset.

1. To find the best configuration of the parameters for each of the models, you must first sample 2% of the data instances. Using only this small subset will keep your computation time manageable. *Hint: if you use Stratified Sampling to ensure a* balance *between positive and negative instances when sampling this random subset of data, you will likely find better parameter values*.

2. You must optimize three configuration parameters each for the Random Forest and GBTs. You may choose any three parameters to optimize for either of the models. You must optimize over a sensible **grid of possible combinations** of values for these three parameters, with at least three possible values for each of these parameters. Use k=3-fold cross-validation to evaluate the performance of each parameter set. Make sure to include the best parameters found for each algorithm in your report. (6 marks)
3. Split the smaller dataset you used for parameter tuning (the 2% of the full dataset you sampled above) into training and test sets. Use the same training-test split to compare the performance of Random Forests (with its best choice of hyperparameters) with that of Gradient Boosted Trees (with its best choice of hyperparameters). Include the performance of each algorithm on both the training and test data sets in your report. (2 marks)

[8 marks]

B.

Once you have found the best parameter configurations for both algorithms using the smaller subset of the data, use the full dataset to compare the performance of Random Forests and Gradient Boosted Trees.

1. Split the full dataset into training and test sets. You will use the same train-test split to evaluate both algorithms. Specify in your report how large the training and test sets are. We will check your code to see that it produces splits of the sizes given in your report. (1 mark)
2. Use the best parameters found for each algorithm (based on the smaller dataset used in the previous step) to fit each model to the training set, and evaluate this model on the test set. In your report, compare performance on both the training and test sets between the two models. (1 mark)

[2 marks]

Question 4

Question 4. Movie Recommendations and Cluster Analysis [set by Robert Loftin - 10 marks]

You should finish the lab session on collaborative filtering before solving part A, and the lab session on clustering before solving part B.

Data:

Use **wget** to download the [MovieLens 20M Dataset]() to the "Data" folder and unzip it there. Please read the [dataset description]() to understand the data before completing the following tasks.

Tasks:

A. Time-split Recommendations [4 marks]:

1. Perform time-split recommendation using ALS-based matrix factorisation in PySpark on the rating data in **ratings.csv**:

- **sort** all data by the timestamp,
- perform **splitting according to the sorted timestamp**. **Earlier times (the past) should be used for training, and later times (the future) should be used for testing**, which is a more realistic setting than random splits. Consider three such splits with training data sizes of 40%, 60%, and 80%.

[1 mark]

For **each** of the three splits above, study two versions (*settings*) of ALS using your student number (keeping only the digits) as the seed for the following

- **Setting 1**: The same ALS setting you used in Lab 6, except for the random seed
- **Setting 2**: Based on the results (*see the next step 3 below*) from the first ALS setting, choose a **different ALS setting that can potentially improve the results.** Provide at least a **one-sentence justification to explain** why you think the chosen setting can potentially improve the results. [*This is to imagine a real scenario. You need to think about how the performance might be improved, provide a justification, and then make changes. This implies that failing to improve the results is* ***acceptable****, but we expect you to provide a good justification when you make changes aiming to improve the results, and that your justification is sound.*]

[2 marks]

2. For each split and each version of ALS, compute three metrics: the Root Mean Square Error (RMSE), Mean Squared Error (MSE), and Mean Absolute Error (MAE). Put these RMSE, MSE and MAE results for each of the three splits into one **Table** for the two ALS settings in the report. You need to report 3 metrics $\times$ 3 splits $\times$ 2 ALS settings = 18 numbers. Visualise these 18 numbers in ONE single figure.

[1 mark]

B. User Analysis [4 marks]:

1) After ALS, **each user** is modelled by a vector of factors. For each of the three time-splits, use *k*-means in PySpark with ***k=25*** to cluster all the users based on the user factors learned with ALS **Setting 2** above, and find the top five largest user clusters. As before, use the digits of your student number as the random seed for initialising the clusters. Report the size of (i.e. the number of users in) each of the top five clusters in one **Table**, in total 3 splits x 5 clusters = 15 numbers. Visualise these 15 numbers in ONE single figure.

[2 marks]

2) For each of the three splits in Q4-A1, consider only the *largest* user cluster in Q4-B1, and do the following on the *training* set only:

- Considering **all users** in the largest user cluster, find all the movies that have been rated by these users and their respective average ratings (over users in this cluster), and name this collection as *movies_largest_cluster*. Find those movies in *movies_largest_cluster* with an average rating greater than or equal to 4 (>=4), and name these as *top_movies*.
- Use **movies.csv** to find the **genres** for all of the *top_movies* and report the top ten most popular genres. *Each movie may have multiple genres, separated by the character* '|'. *Here "most popular" means genres assigned to the largest number of top_movies*). Report these 3 splits x 10 genres = 30 genres in one **Table**.

[2 marks]

C. Discuss your two most interesting observations from A & B above, each with three sentences: 1) What is the observation? 2) What are the possible causes of the observation? 3) How useful is this observation to a movie website such as **Netflix**? Your report must be clearly written, and your code must be well-documented so that it is clear what each step is doing.

[2 marks]

The END of the Assignment

FAQ

FAQ

Q1. Can we use libraries other than PySpark to generate the results?

A1: For functionalities available in PySpark, you should use PySpark, particularly for the core computational part. If functionalities are unavailable in PySpark, you may use other Python libraries.